

Poisson Homework

1. Accidents occur on a certain stretch of motorway at the rate of three per month.

Find the probability that on a given month there will be ...

- a) ... no accidents.
- b) ... one accident.
- c) ... two accidents.
- d) ... three accidents.
- e) ... four accidents.

2. Cretan Airlines services which arrive late to Athens Airport on a typical week can be modelled by a Poisson distribution with mean of 4.5 .

a) Determine the probability that on a given week there will be ...

- a. four late arrivals.
- b. less than four late arrivals.
- c. at least seven late arrivals.

b) Determine the probability that on a given two week period there will be between eight and thirteen (inclusive) late arrivals.

3. A discrete random variable X has Poisson distribution with mean λ .

Given that $P(X = 8) = P(X = 9)$, determine the value of $P(4 < X \leq 10)$.

4. The number of houses sold by an estate agent follows a Poisson distribution, with a mean of 2.5 houses per week.

a) Find the probability that in the next four weeks the estate agent sells ...

- i. ... exactly 4 houses
- ii. ... more than 6 houses.

The estate agent monitors the house sales in periods of 4 weeks.

b) Find the probability that in the next twelve of those four-week periods there are exactly 7 four week periods in which more than 6 houses are sold.

5.

People arrive randomly and independently at the elevator in a block of flats at an average rate of 4 people every 5 minutes.

- (i) Find the probability that exactly two people arrive in a 1-minute period.
- (ii) Find the probability that nobody arrives in a 15-second period.
- (iii) The probability that at least one person arrives in the next t minutes is 0.9. Find the value of t .

6.

It is known that 1.2% of rods made by a certain machine are bent. The random variable X denotes the number of bent rods in a random sample of 400 rods.

- (i) State the distribution of X .
- (ii) State, with a reason, a suitable approximate distribution for X .
- (iii) Use your approximate distribution to find the probability that the sample will include more than 2 bent rods.